

# **Solving Problems Scientifically**

A Comprehensive Technical Framework

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**Gilles Pilon, P.Eng.**



*To TBD*



*"An expert is a person who has made all the mistakes that can be made  
in a very narrow field."* — Niels Bohr



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Avtar Singh, who taught me persistence, drive, and the desire to learn anything.

And  $\text{\LaTeX}$  and  $\text{\LaTeX} 2_{\epsilon}$ .



# Preface

Why did I write this book? TBD



# Note to the Reader

## Work in Progress

Please note that this book is currently a **work in progress**. Content, code snippets, and structural frameworks are actively being developed, refined, and updated. Chapters may be added, reorganized, or revised as the material matures. There is a companion [YouTube channel](#). I do not have a timeline for this work. It is a labour of love.

## Document Conventions

Links are shown in:

- [blue](#) for links to other parts within this book, such as chapters, sections, glossary, and page numbers.
- [red](#) for links to the reference section.
- [orange](#) for links outside the book, such as a website.

Code and shell commands are shown in:

- Shell commands (zsh, bash) are shown inside a box outlined in black, with a note above it that identifies it as shell commands.
- Python commands are shown inside a box outlined in black, with a note above it that identifies it as code.



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# List of Equations

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# Chapter 1

## The technical sandbox

This is a structural sandbox chapter designed to validate typography, math environments, glossary tracking, and custom list indexing. Use this file as a structural blueprint when spawning new chapters.

### 1.1 Create figures with captions

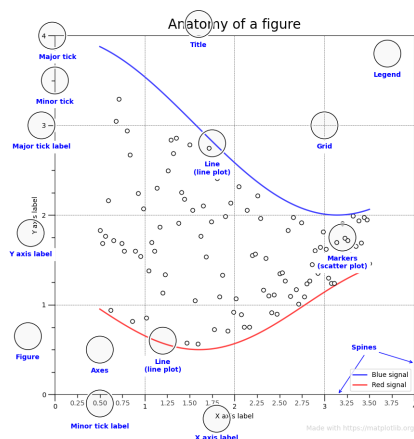


Figure 1.1: Anatomy of a matplotlib figure

### 1.2 Create bold, italics, and underline

Some of the **greatest** *discoveries* in [science](#) were made by *accident*.

### 1.3 Create URLs

Hyperlinks within the document can be easily added. For example, [see Create figures with captions](#) uses `hyperref{}` to take you to a link based on a section or label. And `\ref{fig:labelname}` such as Figure [1.1](#) inserts the figure number as a link to that figure.

Hyperlinks to external URLs can be easily added. For example, see [Wikipedia L<sup>A</sup>T<sub>E</sub>X](#).

## 1.4 Create lists

### 1.4.1 Create unordered lists

- Illinois
  - Chicago
  - Naperville
- Kentucky
  - Lexington
  - Louisville

### 1.4.2 Create ordered lists

1. Cat
  - (a) Persian
  - (b) Siamese
2. Dog
  - (a) Aussiedoodle
  - (b) Newfoundland

## 1.5 Create tables

### 1.5.1 Create simple tables

Table 1.1: Visigothi cum Romanis

|              | I   | II  | III | IV  | V   | VI  |
|--------------|-----|-----|-----|-----|-----|-----|
| Vandali      | 123 | 456 | 678 | 321 | 644 | 768 |
| Visigothorum | 021 | 229 | 678 | 123 | 456 | 678 |

### 1.5.2 Create multi row tables

Table 1.2: Comparison of capability indices

|          | Process requirement                                                                                                                                              | Sampling                                                                                                                                                                                   | Comments                                                                                                                                                                |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $C_{pk}$ | <ul style="list-style-type: none"><li>- Stable and predictable variation</li><li>- Normal distribution</li><li>- Need at least one specification limit</li></ul> | <ul style="list-style-type: none"><li>- Rational</li><li>- Representative of process</li><li>- Ensure same process conditions</li><li>- No identifiable condition changes</li></ul>        | <ul style="list-style-type: none"><li>- Possible cost to increasing <math>C_{pk}</math></li><li>- Wide confidence interval</li></ul>                                    |
| $P_{pk}$ | <ul style="list-style-type: none"><li>- Stable and predictable variation</li><li>- Normal distribution</li><li>- Need at least one specification limit</li></ul> | <ul style="list-style-type: none"><li>- Random</li><li>- Sufficiently large, representative</li><li>- Ensure same process conditions</li><li>- No identifiable condition changes</li></ul> | <ul style="list-style-type: none"><li>- Possible cost to increasing <math>ppk</math></li><li>- Wide confidence interval</li><li>- Easiest formulae to explain</li></ul> |
| $C_{pm}$ | <ul style="list-style-type: none"><li>- Stable and predictable variation</li><li>- Normal distribution</li></ul>                                                 | <ul style="list-style-type: none"><li>- Same as <math>C_{pk}</math> or <math>P_{pk}</math></li></ul>                                                                                       | <ul style="list-style-type: none"><li>- <math>C_{pm}</math> penalizes for being off-target</li><li>- Wide confidence interval</li></ul>                                 |

Table 1.2: Comparison of capability indices

| Process requirement                                                    | Sampling | Comments                             |
|------------------------------------------------------------------------|----------|--------------------------------------|
| - Need at least one specification limit<br>- Need specification target |          | - Most difficult formulae to explain |

## 1.6 Create mathematics

### 1.6.1 Create inline mathematics

Examples of inline mathematics  $\frac{a}{b}$ , followed by text and another inline equation  $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$  and more text  $\sum_{k=0}^{\infty} \frac{1}{2^k} = 2$  and finally much more text to cover many lines of a paragraph.

### 1.6.2 Create display style mathematics

Examples of mathematics on their own lines. All equations are flush-left to the page unless multilines, and thus aligned on a symbol with the longest equation flush-left. The equations numbers are flush-right.

$$\text{measured observation} = \text{true value} + \text{measurement system error} \quad (1.1)$$

$$\text{ICC} = \frac{\text{variance of the product values}}{\text{variance of the product observations}} \quad (1.2)$$

$$\bar{X} \pm \text{CI} = \frac{\sum_{i=1}^n X_i}{n} \pm t_{1-\alpha/2, n-1} \times \frac{s}{\sqrt{n}} \quad (1.3)$$

$$\text{distance to target} = \text{specification target} - \text{location statistic} = T - \bar{X} \quad (1.4)$$

$$s \pm \text{CI} = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}} \pm \left( \sqrt{\frac{(n-1)s^2}{\chi_{\alpha/2}^2}}, \sqrt{\frac{(n-1)s^2}{\chi_{1-\alpha/2}^2}} \right) \quad (1.5)$$

$$\text{distance to nearest spec} = \frac{\text{nearest spec} - \text{location statistic}}{\text{dispersion statistic}} = \frac{\min(\bar{X} - \text{LSL}, \text{USL} - \bar{X})}{s} \quad (1.6)$$

The confidence intervals are:

$$\text{Lower bound} = \hat{P}_{pk} - z_{1-\alpha/2} \sqrt{\frac{1}{\left(\frac{\text{toler}}{2}\right)^2 n} + \frac{(\hat{P}_{pk})^2}{2\nu}} \quad (1.7)$$

$$\text{Upper bound} = \hat{P}_{pk} + z_{1-\alpha/2} \sqrt{\frac{1}{\left(\frac{\text{toler}}{2}\right)^2 n} + \frac{(\hat{P}_{pk})^2}{2\nu}} \quad (1.8)$$

where:

$n$  = the number of observations in the sample

$\nu$  = the degrees of freedom =  $n - 1$

$\chi^2_{\alpha,\nu}$  = the  $\alpha$  percentile of the chi-square distribution with  $\nu$  degrees of freedom

toler = multiplier of the standard deviation tolerance (Minitab® uses 6 as the default value)

$z_{1-\alpha/2}$  = the  $1 - \alpha/2$  percentile of the standard normal distribution

$$A = \begin{pmatrix} 1 & 0 & 4 & -1 \\ 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (1.9)$$

$$A = \begin{bmatrix} 1 & 0 & 4 & -1 \\ 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1.10)$$

## 1.7 Creating greek letters

### 1.7.1 Greek Unicode

Greek letters can be inserted in mathematics and text using Unicode characters. In Neovim, the sequence is 'ControlKey + v uhhhh' where 'hhhh' is the Unicode.

Table 1.3: Greek Unicode

| letter  | code | miniscule | code | majuscule |
|---------|------|-----------|------|-----------|
| alpha   | 03b1 | α         | 0391 | A         |
| beta    | 03b2 | β         | 0392 | B         |
| gamma   | 03b3 | γ         | 0393 | Γ         |
| delta   | 03b4 | δ         | 0394 | Δ         |
| epsilon | 03b5 | ε         | 0395 | E         |
| zeta    | 03b6 | ζ         | 0396 | Z         |
| eta     | 03b7 | η         | 0397 | H         |



Table 1.3: Greek Unicode

| letter  | code | miniscule | code | majuscul |
|---------|------|-----------|------|----------|
| theta   | 03b8 | θ         | 0398 | Θ        |
| iota    | 03b9 | ι         | 0399 | Ι        |
| kappa   | 03ba | κ         | 039a | Κ        |
| lambda  | 03bb | λ         | 039b | Λ        |
| mu      | 03bc | μ         | 039c | Μ        |
| nu      | 03bd | ν         | 039d | Ν        |
| xi      | 03be | ξ         | 039e | Ξ        |
| omicron |      | ο         |      | Ο        |
| pi      | 03c0 | π         | 03a0 | Π        |
| rho     | 03c1 | ρ         | 03a1 | Ρ        |
| sigma   | 03c3 | σ         | 03a3 | Σ        |
| tau     | 03c4 | τ         | 03a4 | Τ        |
| upsilon | 03c5 | υ         | 03a5 | Υ        |
| phi     | 03c6 | φ         | 03a6 | Φ        |
| chi     | 03c7 | χ         | 03a7 | Χ        |
| psi     | 03c8 | ψ         | 03a8 | Ψ        |
| omega   | 03c9 | ω         | 03a9 | Ω        |

## 1.7.2 Greek mathematics

Greek letters can be inserted in mathematics. These are naturally italicized.

Table 1.4: Greek mathematics

| letter  | code                  | miniscule  | code                  | majuscul   |
|---------|-----------------------|------------|-----------------------|------------|
| alpha   | <code>\alpha</code>   | $\alpha$   | A                     | $A$        |
| beta    | <code>\beta</code>    | $\beta$    | B                     | $B$        |
| gamma   | <code>\gamma</code>   | $\gamma$   | <code>\Gamma</code>   | $\Gamma$   |
| delta   | <code>\delta</code>   | $\delta$   | <code>\Delta</code>   | $\Delta$   |
| epsilon | <code>\epsilon</code> | $\epsilon$ | E                     | $E$        |
| zeta    | <code>\zeta</code>    | $\zeta$    | Z                     | $Z$        |
| eta     | <code>\eta</code>     | $\eta$     | H                     | $H$        |
| theta   | <code>\theta</code>   | $\theta$   | <code>\Theta</code>   | $\Theta$   |
| iota    | <code>\iota</code>    | $\iota$    | I                     | $I$        |
| kappa   | <code>\kappa</code>   | $\kappa$   | K                     | $K$        |
| lambda  | <code>\lambda</code>  | $\lambda$  | <code>\Lambda</code>  | $\Lambda$  |
| mu      | <code>\mu</code>      | $\mu$      | M                     | $M$        |
| nu      | <code>\nu</code>      | $\nu$      | N                     | $N$        |
| xi      | <code>\xi</code>      | $\xi$      | <code>\Xi</code>      | $\Xi$      |
| omicron | <code>o</code>        | $o$        | O                     | $O$        |
| pi      | <code>\pi</code>      | $\pi$      | <code>\Pi</code>      | $\Pi$      |
| rho     | <code>\rho</code>     | $\rho$     | P                     | $P$        |
| sigma   | <code>\sigma</code>   | $\sigma$   | <code>\Sigma</code>   | $\Sigma$   |
| tau     | <code>\tau</code>     | $\tau$     | T                     | $T$        |
| upsilon | <code>\upsilon</code> | $\upsilon$ | <code>\Upsilon</code> | $\Upsilon$ |
| phi     | <code>\phi</code>     | $\phi$     | <code>\Phi</code>     | $\Phi$     |
| chi     | <code>\chi</code>     | $\chi$     | X                     | $X$        |
| psi     | <code>\psi</code>     | $\psi$     | <code>\Psi</code>     | $\Psi$     |
| omega   | <code>\omega</code>   | $\omega$   | <code>\Omega</code>   | $\Omega$   |

## 1.7.3 Greek text

Greek letters can be inserted as text.

Table 1.5: Greek text

| letter  | code                      | miniscule | code                      | majuscule |
|---------|---------------------------|-----------|---------------------------|-----------|
| alpha   | <code>\textalpha</code>   | α         | <code>\textAlpha</code>   | A         |
| beta    | <code>\textbeta</code>    | β         | <code>\textBeta</code>    | B         |
| gamma   | <code>\textgamma</code>   | γ         | <code>\textGamma</code>   | Γ         |
| delta   | <code>\textdelta</code>   | δ         | <code>\textDelta</code>   | Δ         |
| epsilon | <code>\textepsilon</code> | ε         | <code>\textEpsilon</code> | E         |
| zeta    | <code>\textzeta</code>    | ζ         | <code>\textZeta</code>    | Z         |
| eta     | <code>\texteta</code>     | η         | <code>\textEta</code>     | H         |
| theta   | <code>\texttheta</code>   | θ         | <code>\textTheta</code>   | Θ         |
| iota    | <code>\textiota</code>    | ι         | <code>\textIota</code>    | I         |
| kappa   | <code>\textkappa</code>   | κ         | <code>\textKappa</code>   | K         |
| lambda  | <code>\textlambda</code>  | λ         | <code>\textLambda</code>  | Λ         |
| mu      | <code>\textmu</code>      | μ         | <code>\textMu</code>      | M         |
| nu      | <code>\textnu</code>      | ν         | <code>\textNu</code>      | N         |
| xi      | <code>\textxi</code>      | ξ         | <code>\textXi</code>      | Ξ         |
| omicron | <code>\textomicron</code> | ο         | <code>\textO</code>       | O         |
| pi      | <code>\textpi</code>      | π         | <code>\textPi</code>      | Π         |
| rho     | <code>\textrho</code>     | ρ         | <code>\textRho</code>     | Ρ         |
| sigma   | <code>\textsigma</code>   | σ         | <code>\textSigma</code>   | Σ         |
| tau     | <code>\texttau</code>     | τ         | <code>\textTau</code>     | T         |
| upsilon | <code>\textupsilon</code> | υ         | <code>\textUpsilon</code> | Υ         |
| phi     | <code>\textphi</code>     | φ         | <code>\textPhi</code>     | Φ         |
| chi     | <code>\textchi</code>     | χ         | <code>\textChi</code>     | Χ         |
| psi     | <code>\textpsi</code>     | ψ         | <code>\textPsi</code>     | Ψ         |
| omega   | <code>\textomega</code>   | ω         | <code>\textOmega</code>   | Ω         |

## 1.8 Inserting citations

We also anchor technical references using our external database system (Wheeler and Lyday 1989). Here is a single citation without prenote and postnote. There is a definitive book on MSA (Wheeler 2006). Here is a single citation with prenote only. There is a definitive book on SPC (Wheeler 2006). Here is a single citation with postnote only.  $C_{pm}$  is a complex subject (Chan, Cheng, and Spiring 1988, pp. 1–10).

## 1.9 Inserting glossary terms

For glossary terms use [endogenous variable](#), [endogenous variables](#), [exogenous variable](#), [exogenous variables](#), [machine learning](#), and [seborrheic keratosis](#).

## 1.10 Inserting index terms

The development of advanced automation frameworks requires a tight integration between custom infrastructure and robust software engineering. When compiling highly optimized modules, developers must pay close attention to memory constraints.

## 1.11 Endnotes

The initial deployment of machine learning architectures required massive mainframe hardware systems.<sup>1</sup> This structural limitation heavily restricted real-time calculations.

# Chapter 2

## Introduction

Write your introductory paragraphs directly here.

### 2.1 Background and Context

Type your background details straight into this block. If you have an equation, write it here just like you did in the sandbox.

### 2.2 Project Objectives

List your main goals here.

The development of advanced automation frameworks requires a tight integration between custom infrastructure and robust software engineering. When compiling highly optimized modules, developers must pay close attention to memory constraints.



## Chapter 3

# Scientific method

The scientific method is doing something in a systematic and logical manner. A claim is verified through rigorous trails, not anecdotal evidence.

From one source:

- observation
- hypothesis
- experimentation
- conclusion

Other thoughts:

- what problem are you trying solve?
- establish methods of monitoring and control

What does Wikipedia say?

What does the Cambridge Dictionary say?



## **Appendix A**

# **Appendix—Python**

This appendix will show how to install and use Python.

### **A.1 Install Python**

TBD





## Appendix B

# Appendix—dawgdad

### B.1 Install dawgdad

**dawgdad** (Pilon 2026) is a **Python** package for statistical, graphical, and predictive analysis.

dawgdad requires Python. Install a **Python** distribution appropriate to your operating system (**Appendix—Python**).

Install dawgdad following the instructions at <https://github.com/gillespilon/dawgdad>. dawgdad is compatible with the **latest Python distribution** from **Python.org**. Instructions on how to use dawgdad are provided in relevant sections of this document. Full documentation is available at <https://dawgdad.readthedocs.io/en/latest/>.

Run the following command in your shell:

```
$ Shell: Installing dawgdad
```

```
pip install dawgdad
```

If you prefer a different distribution, such as **Anaconda**, following the instruction as appropriate.

### B.2 Using dawgdad in Python

Here is a quick example of how to import and initialize the module:

```
▷ Code: Initializing the data framework
```

```
1 import dawgdad as dd
2
3 # Load and analyze your dataset
4 data = dd.load_analytics("performance_log.csv")
5 print(data.summary())
```



# Glossary

## A

**average** It is the sum of a set of values divided by the count of values. It is a type of [location statistic](#).

$$\bar{Y} = \frac{\sum_{i=1}^n X_i}{n}$$

[16](#)

## C

**computer program** A computer program is a sequence of instructions written in a [programming language](#) that is executed by a computer. [16](#)

## D

**data set** A data set is a collection of [data](#). In the case of tabular [data](#), a data set corresponds to one or more database tables, where every column of a table represents a particular variable, and each row corresponds to a given record of the data set in question. The data set lists values for each of the variables, such as for example height and weight of an object, for each member of the data set. Data sets can also consist of a collection of documents or files. [16, 17](#)

**datum** A datum is an individual value in a collection of data, discrete or continuous values that convey information, describing the quantity, quality, fact, [statistic](#), other basic units of meaning, or simply sequences of symbols that may be further interpreted formally. Data are usually organized into structures such as tables that provide additional context and meaning, and may themselves be used as data in larger structures. [15-17](#)

**designed experiment** It is a method to change factors simultaneously (inputs) using a structured pattern, in order to determine their effect on some response(s) (output(s)). [15](#)

## E

**endogenous variable** Endogenous variables are explained or predicted by a model. They are caused by factors within the system. Other names are y, dependent variable, regressand, outcome, response variable, and target ([machine learning](#)). In  $y = mX + b$ , y is the endogenous variable. [6, 17](#)

**exogenous variable** Exogenous variables are terms in a model. They are caused by factors outside the system. Other names are x, independent variable, regressor, design, explanatory variable, and feature ([machine learning](#)). In  $y = mX + b$ , x is the exogenous variable. In a [designed experiment](#), it is a factor that you can control, want to control, or is controlled at low cost. [6, 17](#)

**experiment** It is a series of [observations](#) conducted under controlled conditions to study a relationship with the purpose of drawing causal inferences about that relationship. An experiment involves the manipulation of an independent variable, the measurement of a dependent variable, and the exposure of various participants to one or more of the conditions being studied. Random selection of participants and their random assignment to conditions also are necessary in experiments. [16](#)

## L

**labeled data** Labeled [data](#) is a collection of raw [data](#) that has been assigned labels to provide context and meaning. Labels can be used to categorize [data](#), such as by indicating whether an image contains a cat or dog, or if an audio recording includes a specific word. [16](#)

**location statistic** It is a statistic that estimates the central value that best describes the [data](#). Examples include the [average](#), [median](#), and [mode](#). [15](#)

## M

**machine learning** It is a field of study that gives computers the ability to learn without being explicitly programmed ((Samuel [1959](#))). It is an approach to making repeated decisions that involves algorithmically finding patterns in [data](#) and using these to make models that deal correctly with new [data](#). There are four types of machine learning: [supervised machine learning](#), [unsupervised machine learning](#), [semi-supervised learning](#), and [reinforcement learning](#). A common task in machine learning is to build a model from [data sets](#). There are three [data sets](#) commonly used in the different stages of the creation of the model: [training data set](#), [validation data set](#), and [test data set](#). [6](#), [15–17](#)

**machine learning hyperparameter** In [machine learning](#) it is a [machine learning parameter](#) that can be set in order to define any configurable part of a model's learning process. Hyperparameters can be classified as either model hyperparameters or algorithm hyperparameters. [17](#)

**machine learning parameter** In [machine learning](#) it is a characteristic that the model learns from the [data set](#). [16](#)

**median** It is the value of the point which has half the [data](#) smaller than that point and half the [data](#) larger than that point. There are various formulae to calculate the median, for example JMP, Minitab, and SAS estimate it in different ways. [16](#)

**mode** It is the value of the random sample that occurs with the greatest frequency. [16](#)

## O

**observation** It is the careful, close examination of an object, process, or other phenomenon for the purpose of collecting data about it or drawing conclusions. [15](#)

## P

**probability distribution** In [probability theory](#) and [statistics](#), a probability distribution is the mathematical function that gives the probabilities of occurrence of possible outcomes for an [experiment](#). It is a mathematical description of a random phenomenon in terms of its [sample space](#) and the probabilities of event (subsets of the [sample space](#)). [17](#)

**probability theory** TBD [16](#)

**programming language** A programming language is a system of notation for writing [computer programs](#). [15](#), [16](#)

**Python** Python is an easy to learn, powerful [programming language](#). It has efficient high-level [data](#) structures and a simple but effective approach to object-oriented programming. Python's elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms. ((Python Software Foundation [2024](#))) [13](#)

## R

**reinforcement learning** It is an area of [machine learning](#) concerned with how intelligent agents ought to take actions in an environment in order to maximize the notion of cumulative reward. [16](#)

## S

**sample space** It is the set of all possible outcomes of a random phenomenon being observed. [16](#)

**science** Science is a systematic discipline that builds and organizes knowledge in the form of testable hypotheses and predictions about the universe. It establishes objective truths, not by wishful or magical thinking, but by repeated and verified experiments. This makes such truths true, whether or not you believe in them. [1](#)

**seborrheic keratosis** It occurs when skin cells, keratinocytes, multiply rapidly, resulting in a non-cancerous growth. This slow-growing spot is typically raised and may have a rough texture, and can appear anywhere on your body. It is neither viral or bacterial and cannot spread. It is most common in adults over 50 years of age. See [Yale Medicine—Seborrheic Karatosis](#). [6](#)

**semi-supervised learning** It is an approach to [machine learning](#) that combines a small amount of [labeled data](#) to guide the model with a large amount of [unlabeled data data](#) during training. Semi-supervised learning falls between unsupervised learning (with no labeled [training data set](#)) and supervised learning (with only labeled [training data set](#)). It is a special instance of weak supervision. [16](#)

**statistic** TBD [15](#)

**statistics** TBD [16](#)

**supervised machine learning** It is the [machine learning](#) task of learning a function that maps an input to an output based on example input-output pairs. The five steps of supervised learning are: 1. import the classes you

plan to use, 2. instantiate the transformers, classifiers, regressors, and selectors, 3. train the model on an existing [data set](#), 4. tune the [machine learning hyperparameters](#), and 5. predict the response for a new [data set](#). 16

## T

**test data set** It is a [data set](#) that is independent of the [training data set](#) but follows the same [probability distribution](#) as the [training data set](#). It is used to assess the performance of the [endogenous variable](#). 16

**training data set** It is a [data set](#) of examples used during the learning process and is used to fit the [exogenous variables](#) to the [endogenous variables](#). 16, 17

## U

**unlabeled data** Unlabeled [data](#) is raw [data](#) that doesn't have any explicit tags or categories. It's a collection of samples of items, such as images, videos, articles, or social media posts, that are not labeled or explained. 16, 17

**unsupervised machine learning** Unsupervised machine learning is a type of [machine learning](#) that uses algorithms to analyze [unlabeled data data](#) and discover patterns without human intervention. In unsupervised learning, a model is given input [data](#) and is allowed to identify patterns and relationships on its own without any explicit labels or goals. 16

## V

**validation data set** It is a [data set](#) of examples used during the learning process to tune the [machine learning hyperparameters](#) of a model. 16



# Endnotes

## Notes

### Notes for Chapter 1

1. See Smith (1982) for an architectural breakdown of early mainframe limitations.





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When not structuring complex codebases or optimizing technical documentation frameworks, he is an avid cyclist focused on performance analytics and outdoor endurance. He lives with his family, continuing to build open-source tools and data utilities.



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